

Recursive Bipartite Matching Distance for Game Tree Abstraction with Bounded Error

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Abstract

We introduce the *recursive bipartite matching* (RBM) distance, a metric on rooted trees defined by recursively solving minimum-cost bipartite matchings over children. Unlike scalar hand-strength metrics used in prior game abstraction work, RBM captures the full strategic structure of continuation trees. We prove three results: (1) RBM is a metric when the leaf distance is a metric; (2) merging two trees at distance ε introduces expected-value error at most $\varepsilon/2$ under chance-uniform leaf averaging (a leaf-mean bound, not yet a Nash exploitability bound; we discuss the bridge to exploitability via Kroer and Sandholm [2014, 2018] in Section 3); and (3) an online learner that clusters game states by RBM distance achieves cumulative regret $O(KV_{\max} + T\varepsilon/2)$ with fixed threshold, or $O(\sqrt{T})$ under ε -annealing. On Rhode Island Hold'em, RBM-based abstraction has lower EV error than Earth Mover's Distance abstraction across all non-trivial compression levels we evaluate. On full 2-player Limit Hold'em with a standard 52-card deck—the benchmark game solved by Bowling et al. [2015]—RBM has lower preflop EV error than EMD across all five compression levels we measure, with up to 42% less error. In a 20,000-hand head-to-head match between MCCFR bots trained on the two abstractions, the RBM bot finishes +0.02 bb/hand ahead, but the result is not statistically significant (95% CI $\approx [-0.19, +0.23]$). On No-Limit Hold'em, RBM wins abstraction quality at all five compression levels at 20bb (up to 73% less error). At 200bb, where pairwise RBM distance is computed on showdown-distribution trees rather than full game trees, RBM dominates fine-grained compression (60% less error at $k=25$, 40% at $k=15$) but loses at coarser levels.

1 Introduction

Abstraction is the dominant paradigm for solving large extensive-form games. The idea is straightforward: collapse strategically similar game states into equivalence classes, solve the smaller *abstract* game, and map the abstract strategy back to the original. Superhuman poker agents [Bowling et al., 2015, Moravčík et al., 2017, Brown and Sandholm, 2018] all rely on some form of abstraction to make computation tractable.

The quality of an abstraction depends critically on the *distance metric* used to decide which states are “similar.” The standard approach in poker AI, pioneered by Gilpin and Sandholm [2007], clusters information sets using Earth Mover's Distance (EMD) on hand-strength histograms: distributions over scalar equity values obtained by evaluating each hand against a uniform random opponent. This is effective but inherently limited—it reduces each game state to a one-dimensional summary, discarding the branching structure of future play.

Two hands may have identical equity distributions yet lead to fundamentally different strategic situations. A made hand and a draw can have the same expected value but completely different

*Work in progress. Code: <https://github.com/structured-labs/recursive-bipartite-matching>

continuation trees: one calls safely, the other bets aggressively to deny correct pot odds or folds against reraises. Scalar metrics cannot distinguish them.

We propose a structural alternative. The *recursive bipartite matching* (RBM) distance between two game trees is defined by recursively computing minimum-cost bipartite matchings between their children at every level. Leaves are compared by payoff difference; internal nodes are compared by the cost of optimally aligning their subtrees. Structural mismatches (differing numbers of children) incur an explicit phantom penalty. The result is a metric that respects both the values at the leaves and the *shape* of the computation that produces them.

This paper develops the theory and practice of RBM-based game abstraction in three parts.

The metric and its error bound. We prove that RBM is a metric (Section 2), and that merging two trees at distance ε introduces EV error at most $\varepsilon/2$ (Section 3). The bound is tight at leaves but strictly tighter at branching nodes, where averaging reduces error.

The location problem. Using a structurally rich distance creates a compressed *EV graph*—a DAG of merged game states—that faithfully preserves strategic content. But locating oneself in this graph during live play requires comparing one’s continuation tree against cluster representatives, which can be as expensive as solving the uncompressed game (Section 4).

The online learning resolution. We dissolve the location problem by building the EV graph *incrementally through self-play*. The learner always knows where it is because it constructed the graph from its own positions. We prove that this online learner achieves bounded cumulative regret, with sublinear average regret under ε -annealing (Section 5).

Experiments on Rhode Island Hold’em [Gilpin and Sandholm, 2005], full 2-player Limit Hold’em [Bowling et al., 2015], and No-Limit Hold’em [Brown and Sandholm, 2018] (Section 6) report consistent improvements in abstraction quality: RBM-based clustering produces lower EV error than EMD across all non-trivial compression levels we evaluate, on Rhode Island Hold’em, on Limit Hold’em (5–0 at $k \in \{3, 5, 10, 15, 25\}$, up to 42% less error), and on No-Limit 20bb (up to 73% less error at $k=25$). At 200bb, RBM dominates only at fine-grained compression ($k \geq 15$) and loses at coarser levels—an asymmetry consistent with the abstraction-pathology phenomenon documented by Waugh et al. [2009] and amplified here because the 200bb experiment uses showdown-distribution trees as a proxy for the full 186,174-node game tree (full-tree RBM at 200bb remains future work, Section 8). MCCFR bots trained on RBM versus EMD abstractions in Limit Hold’em finish within statistical noise of each other in head-to-head play (RBM +0.02 bb/h, $n=20,000$, 95% CI $\approx [-0.19, +0.23]$): no detectable gameplay difference, despite the abstraction-quality gap. The online learner achieves 99.4% cache hit rates with ~ 400 naturally emerging postflop clusters.

2 Recursive Bipartite Matching Distance

2.1 Definition

Let $\mathcal{T}$ denote the space of finite rooted trees with real-valued leaves. We define a distance $d : \mathcal{T} \times \mathcal{T} \rightarrow \mathbb{R}_{\geq 0}$ recursively.

Definition 2.1 (RBM Distance). *Given two rooted trees T_1 and T_2 :*

Base case. *If both are leaves with values v_1, v_2 :*

$$d(\text{leaf}_1, \text{leaf}_2) = |v_1 - v_2|.$$

Recursive case. Let n_1 have children $\{c_1^1, \dots, c_1^m\}$ and n_2 have children $\{c_2^1, \dots, c_2^n\}$, with $m \leq n$ (WLOG).

1. Compute the pairwise cost matrix W where $W[i, j] = d(\text{sub}(c_1^i), \text{sub}(c_2^j))$.
2. Pad the smaller child set with $|n - m|$ phantom nodes. The cost of matching a real subtree S against a phantom is $\delta(S)$.
3. Solve the minimum-cost perfect matching on the padded bipartite graph via the Hungarian method [Kuhn, 1955], yielding optimal matching M^* .
4. $d(T_1, T_2) = \text{cost}(M^*)$.

The phantom penalty δ quantifies the cost of structural mismatch. The cleanest formulation is $\delta(S) = d(S, \emptyset)$, treating the phantom as a designated empty tree in the metric space. Under this choice, all conditions needed for the metric proof are consequences of the triangle inequality at lower depth (see Remark 2.3). To make this rigorous we extend the base case of Definition 2.1 to cover $\emptyset$ explicitly: $d(\text{leaf}(v), \emptyset) = |v|$ and $d(T, \emptyset) = \sum_{c \in \text{children}(T)} d(\text{sub}(c), \emptyset)$ when T is internal. With this completion, δ is well-defined for every tree, and Remark 2.3 below recovers the 1-Lipschitz and phantom-triangle properties as instances of the lower-depth triangle inequality.

Children are treated as *unordered sets*. The matching finds the best structural alignment without requiring a canonical ordering—exactly the right semantics for game trees, where children of a chance node (possible deals) or decision node (possible actions) have no inherent order.

2.2 Metric Properties

Theorem 2.2 (RBM is a Metric). *Let d_L be a metric on leaf values and let $\delta(S) = d(S, \emptyset)$ for a designated empty tree $\emptyset$. Then the RBM distance d is a metric on $\mathcal{T}$.*

Proof sketch. By structural induction on tree depth. The base case ($k = 0$) reduces to d_L , which is a metric by assumption.

Identity: The identity matching $\{(c_i, c_i)\}$ has cost zero by induction; since all costs are nonneg, the minimum-cost matching achieves $d(T, T) = 0$.

Symmetry: The cost matrix for $d(T_2, T_1)$ is the transpose of that for $d(T_1, T_2)$, using symmetry at depth $< k$. The minimum-cost perfect matching is invariant under transposition.

Triangle inequality: Given optimal matchings M_{12} (between T_1, T_2) and M_{23} (between T_2, T_3), compose through T_2 to construct a feasible matching M_{13} between T_1 and T_3 . Children “through-matched” via some $b_j \in T_2$ satisfy $d(\text{sub}(a_i), \text{sub}(c_k)) \leq d(\text{sub}(a_i), \text{sub}(b_j)) + d(\text{sub}(b_j), \text{sub}(c_k))$ by induction. Dangling children (matched to phantoms on one side) are handled by the 1-Lipschitz property of δ . Summing over all pairs: $\text{cost}(M_{13}) \leq \text{cost}(M_{12}) + \text{cost}(M_{23})$. Since $d(T_1, T_3) \leq \text{cost}(M_{13})$ by optimality, the triangle inequality follows.

The full proof appears in Appendix A. □

Remark 2.3 (Phantom Penalty). *With $\delta(S) = d(S, \emptyset)$, the phantom triangle property $d(S_1, S_2) \leq \delta(S_1) + \delta(S_2)$ and the 1-Lipschitz condition $\delta(S_1) \leq d(S_1, S_2) + \delta(S_2)$ are both instances of the triangle inequality. The theorem is self-contained: if the leaf distance is a metric, then RBM with empty-tree phantoms is a metric.*

2.3 Relation to Prior Tree Distances

RBM is related to but distinct from tree edit distance [Zhang, 1996], which uses insert/delete/relabel operations on ordered trees. Edit distance is more flexible but less natural for *unordered* trees where children are interchangeable. The recursive matching structure also connects to hierarchical

optimal transport [Mémoli, 2011] and the Tree Mover’s Distance [Chuang et al., 2022], but those works target different applications (shape matching and GNN analysis, respectively). See Section 7 for a detailed comparison.

3 Game Tree Compression via Merge

3.1 The Merge Operation

The RBM distance induces a natural *merge* operation. Given trees T_1, T_2 with small distance, produce a representative T^* that captures their shared structure.

Definition 3.1 (Tree Merge). *Given the optimal matching M^* between children of T_1 and T_2 :*

- **Matched pairs** $(c_1^i, c_2^j) \in M^*$: *recursively merge their subtrees.*
- **Phantom-matched children**: *either retain at reduced weight or drop (lossy compression).*
- **Leaf merging**: *average the payoff values: $v^* = (v_1 + v_2)/2$.*

3.2 The EV Graph

Repeated merging across all subtrees at each depth produces a directed acyclic graph—the *EV graph*. Multiple original game paths converge to the same node, meaning “from here, these situations play out essentially the same way.” This is hierarchical clustering over subtrees using RBM as the metric, with each cluster represented by a merged prototype tree.

3.3 EV Error Bound

Theorem 3.2 (EV Error Bound Under Merging). *Let T_1, T_2 be rooted trees with real-valued leaves, $d(T_1, T_2) = \varepsilon$. Let $T^* = \text{merge}(T_1, T_2)$ with equal-weight leaf averaging. Then:*

$$|\text{EV}(T^*) - \text{EV}(T_i)| \leq \frac{\varepsilon}{2}, \quad i \in \{1, 2\}.$$

Proof sketch. By structural induction on tree depth.

Base case (leaves): $T^* = \text{leaf}((v_1 + v_2)/2)$, so $|\text{EV}(T^*) - v_1| = |v_2 - v_1|/2 = \varepsilon/2$.

Inductive case: Let M^* be the optimal matching with matched-pair costs d_i and total $\varepsilon = \sum_i d_i + \sum_{\text{phantom}} \delta(b_j)$. Each merged child satisfies $|\text{EV}(c_i^*) - \text{EV}(\text{sub}(a_i))| \leq d_i/2$ by induction. With uniform weighting over m children:

$$|\text{EV}(T^*) - \text{EV}(T_1)| \leq \frac{1}{m} \sum_{i=1}^m \frac{d_i}{2} = \frac{1}{2m} \sum_{i=1}^m d_i \leq \frac{\varepsilon}{2}.$$

For trees with branching factor $m > 1$ and no phantom terms, the bound tightens to $\varepsilon/(2m) \leq \varepsilon/4$. The stated bound $\varepsilon/2$ is tight only at leaves or degenerate single-child chains. $\square$

Remark 3.3 (EV bound vs. exploitability). *Theorem 3.2 bounds $|\text{EV}(T^*) - \text{EV}(T_i)|$, where EV is the expectation over the tree’s leaves under the chance distribution already baked into the leaf weights—the EV that a fixed strategy profile delivers when one substitutes the merged tree for the original. This is not the same as a bound on Nash exploitability, which asks how much a best-responder can win against a strategy obtained by solving the abstract game and lifting back to the full game. The canonical exploitability bounds for game-tree abstractions are due to Kroer and Sandholm [2014, 2018]; a tight bridge from RBM-induced merge error to exploitability would proceed*

via their level-by-level reward-and-action distance machinery, applied to the specific child-matching structure RBM produces. We treat that bridge as the natural follow-up; the present bound should be read as a necessary but not sufficient condition for low exploitability.

Corollary 3.4 (Weighted Merges). *For unequal-weight merges with weights $w_1 + w_2 = 1$:*

$$|\text{EV}(T^*) - \text{EV}(T_i)| \leq w_{3-i} \cdot \varepsilon.$$

Equal weights $w_1 = w_2 = 1/2$ recover $\varepsilon/2$.

Corollary 3.5 (Iterated Merges). *When k trees $T_1, \dots, T_k$ are merged into a single T^* via successive pairwise merges:*

$$|\text{EV}(T^*) - \text{EV}(T_i)| \leq \max_j d(T_i, T_j) \leq \varepsilon_{\text{diam}},$$

where $\varepsilon_{\text{diam}}$ is the cluster diameter.

4 The Location Problem

Building the EV graph offline is one challenge; *finding yourself in it during live play* is another.

4.1 Statement

Given a compressed EV graph G and a game state v (known: cards, history), find the corresponding node $\phi(v) \in G$.

In flat abstraction, location is trivial: precompute a lookup table mapping each state to its bucket, $O(1)$ at query time. With tree-based abstraction, the equivalence class of v depends on its *entire continuation subtree*—the same non-local structure that makes the distance expressive.

4.2 Complexity Barrier

Proposition 4.1 (Location Cost). *Let b be the branching factor, h the remaining depth, K the number of clusters. Naïve location requires $O(K \cdot b^{2h})$ time per game step, since each distance computation $T(h)$ satisfies $T(h) = b^2 \cdot T(h-1) + O(b^3)$, giving $T(h) = O(b^{2h})$.*

Over a full game of d steps, the total cost is $O(d \cdot K \cdot n^2)$ where $n = b^h$ is the number of leaves. This can exceed the cost of solving the uncompressed game, defeating the purpose of compression. The location problem likely admits a lower bound of $\Omega(n)$ per step, because the equivalence classes depend on global subtree structure defined by optimal (Hungarian) matchings.

This reveals a fundamental tradeoff in game abstraction: *the more structurally faithful the abstraction, the harder it is to navigate at runtime.*

5 Online Learning with Bounded Regret

The offline approach—enumerate, cluster, then look up—hits the location problem wall. We propose a more natural formulation: build the EV graph *dynamically through play*.

5.1 The Online EV Graph Learner

Algorithm 1 resolves the core tensions:

Algorithm 1 Online EV Graph Construction via Self-Play

```
1: Initialize EV graph  $G \leftarrow \emptyset$ , threshold  $\varepsilon > 0$ 
2: for each game  $t = 1, 2, \dots, T$  do
3:   for each decision point in game  $t$  do
4:     Observe game state  $s$ ; build partial subtree  $T_s$ 
5:     Find nearest cluster:  $c^* = \arg \min_c d(T_s, \text{rep}(c))$ 
6:     if  $d(T_s, \text{rep}(c^*)) \leq \varepsilon$  (cache hit) then
7:       Use cluster  $c^*$ 's strategy; record outcome
8:     else {(cache miss)}
9:       Create new cluster from  $T_s$ ; explore
10:    end if
11:  end for
12:  Update cluster representatives with observed payoffs
13:  Merge clusters where  $d(\text{rep}(c_i), \text{rep}(c_j)) < \varepsilon_{\text{merge}}$ 
14:  Optionally anneal  $\varepsilon$  downward
15: end for
```

The location problem dissolves. The learner never needs to locate itself in a precomputed graph because it built the graph *from* its own positions. Each cluster knows its game states because it was born from one.

The enumeration problem dissolves. Only states actually encountered are represented. The graph concentrates resolution where it matters—the same insight behind MCTS [Coulom, 2006].

The error bound becomes a regret bound. When the learner uses cluster c^* 's strategy on a state at distance $\leq \varepsilon$, the EV loss is bounded by $\varepsilon/2$ (Theorem 3.2).

5.2 Regret Bound

Theorem 5.1 (Online Regret Bound). *Consider the online EV graph learner (Algorithm 1) with threshold $\varepsilon > 0$, playing T games against a stationary environment. Let K be the number of distinct strategic equivalence classes (game trees up to ε -distance), and $V_{\max} = \max_t |\text{EV}(T_t)|$. Then:*

1. **Exploration bound:** Total cache misses $\leq K$.
2. **Per-step exploitation error:** At most $\varepsilon/2$ per cache hit.
3. **Cumulative regret:**

$$R(T) \leq K \cdot V_{\max} + (T - K) \cdot \frac{\varepsilon}{2}.$$

4. **Average regret:** $R(T)/T \rightarrow \varepsilon/2$ as $T \rightarrow \infty$.
5. **ε -annealing:** With $\varepsilon(t) = \varepsilon_0/\sqrt{t}$, $R(T) = O(\sqrt{T})$ and $R(T)/T \rightarrow 0$.

Proof sketch. Part 1: Each equivalence class triggers at most one cache miss. Once representative T_t is created, all future trees in the same class satisfy $d(T_{t'}, T_t) \leq \varepsilon$, producing cache hits.

Parts 2–3: Cache hits incur error $\leq \varepsilon/2$ (Theorem 3.2). Cache misses incur at most $V_{\max}$. Summing: $R(T) \leq KV_{\max} + (T - K)\varepsilon/2$.

Part 5: With $\varepsilon(t) = \varepsilon_0/\sqrt{t}$, exploitation regret sums to $\sum_{t=1}^T \varepsilon_0/(2\sqrt{t}) \leq \varepsilon_0\sqrt{T}$. Exploration cost is at most $N \cdot V_{\max}$ (constant). Total: $R(T) = O(\sqrt{T})$.

The full proof appears in Appendix C. □

5.3 Connection to Lipschitz Bandits

The $O(\sqrt{T})$ rate matches the minimax-optimal rate for multi-armed bandits [Auer et al., 2002]. The structural advantage over generic bandits is that RBM provides a *geometry* on the arm space: nearby clusters have similar payoffs by the EV error bound. This is analogous to Lipschitz bandits or continuum-armed bandits [Kleinberg, 2005], where metric structure enables faster learning. In the EV graph setting, the metric is not assumed but *derived* from the game structure via recursive bipartite matching.

6 Experiments

We evaluate on three domains: (1) Rhode Island Hold’em [Gilpin and Sandholm, 2005], a simplified poker variant (2 players, 1 hole card, 2 community cards, 3 betting rounds, limit betting), (2) full 2-player Limit Hold’em with a standard 52-card deck—the benchmark game solved by Bowling et al. [2015], and (3) No-Limit Hold’em (heads-up, 20bb stacks)—the game variant underlying modern superhuman agents [Brown and Sandholm, 2018].

The implementation is in OCaml ($\sim 15\text{K}$ lines) using Jane Street Core libraries, with an $O(n^3)$ Hungarian algorithm for bipartite matching.

6.1 RBM vs EMD Comparison

We compare three distance metrics for game abstraction:

- **RBM**: recursive bipartite matching on full game trees (~ 1141 nodes per deal),
- **EMD**: Earth Mover’s Distance on showdown outcome distributions (win/draw/lose probabilities), the standard poker AI metric [Gilpin and Sandholm, 2007],
- **Scalar EV**: absolute difference in expected values.

For each metric, we compute pairwise distances on 25 sampled deals (3-rank deck, fixed P1 card, varied opponent/community), then perform agglomerative clustering at multiple ε thresholds. At each compression level (cluster count), we measure the maximum EV error across clusters.

Table 1 shows that on this 3-rank toy RBM produces zero EV error at every compression level $k \geq 2$, while EMD’s error sits at the 30.28 floor described above. This is mostly a statement about EMD’s failure mode on a deck this small: with so few distinct showdown histograms, the metric simply cannot resolve hands that have identical showdown distributions but different game-tree continuations. RBM, operating on the full game tree, separates these by construction. We include the table to illustrate the failure mode—not as evidence about which metric is better at production scale.

6.2 CFR on Compressed Games

We run counterfactual regret minimization (CFR) [Zinkevich et al., 2007] on both the full game and RBM-compressed abstractions to measure exploitability—the gap between the computed strategy and Nash equilibrium.

Table 2 shows that on this 3-rank, 10-info-set toy, RBM compression to 3 clusters preserves zero exploitability across the ε range we tested. The natural reading is that RBM has correctly identified the three rank-vs-board equivalence classes the game admits at this scale; we are reluctant to call the result “strategically lossless” in any general sense, because the result also depends on the unique structure of the 3-rank game (with only 10 information sets, there is little room for an abstraction to do damage). Only at $\varepsilon = 1000$, where the algorithm collapses all hands into one

Table 1: RBM vs EMD vs Scalar EV: maximum EV error at matched cluster counts (3-rank Rhode Island Hold'em, 25 sampled deals with varied community cards). Lower is better. RBM achieves zero error down to $k = 3$ clusters while EMD has 30.28 error at every $k \geq 3$. The constant 30.28 reflects a degeneracy of EMD on this 3-rank toy: the showdown-distribution histograms collapse to a small set of equity values, so refining the cluster count cannot reduce error past a floor. We therefore treat this table as illustrative of the representational ceiling of scalar metrics on toy decks, not as a benchmark; the more informative comparisons are the 52-card abstraction-quality results in Sections 6.7 and 6.10.

Clusters	Compress	RBM Error	EMD Error	EV Error	Winner
25	1.0×	0.00	30.28	30.28	RBM
15	1.7×	0.00	30.28	30.28	RBM
10	2.5×	0.00	30.28	30.28	RBM
7	3.6×	0.00	30.28	30.28	RBM
5	5.0×	0.00	30.28	30.28	RBM
3	8.3×	0.00	30.28	30.28	RBM
2	12.5×	25.96	34.07	34.07	RBM
1	25.0×	54.51	54.51	54.51	tie

Table 2: CFR exploitability on full vs. RBM-compressed Rhode Island Hold'em (3-rank, 2 betting rounds, 1000 CFR iterations). RBM compression to 3 clusters achieves **zero exploitability**—the abstraction is strategically lossless.

ε	Clusters	EV Error	Exploitability
(full)	10	—	0.0000
0	3	0.0000	0.0000
50	3	0.0000	0.0000
100	3	0.0000	0.0000
200	3	0.0000	0.0000
500	3	0.0000	0.0000
1000	1	7.0926	5.6573

cluster, does exploitability rise—consistent with Theorem 3.2’s leaf-mean bound but not, on its own, evidence of a tight exploitability bound at larger scales (cf. Remark 3.3).

6.3 Multi-Scale Results

Table 3 shows RBM vs EMD across three game scales. RBM dominates at every scale, achieving zero error at aggressive compression levels where EMD retains substantial error.

6.4 Metric Property Verification

We verify RBM metric properties exhaustively on the 4-rank game (91 deal trees, 753,571 triples for the triangle inequality):

- **Identity:** $d(T, T) = 0$ for all 91 trees. ✓
- **Symmetry:** $d(T_i, T_j) = d(T_j, T_i)$ for all 4095 pairs. ✓
- **Triangle inequality:** verified on all 753,571 triples. ✓

Table 3: RBM vs EMD across game scales (30 sampled deals per scale, varied community cards). RBM wins column counts victories across 8 target compression levels. RBM achieves zero error down to $k = 2$ at all scales.

Scale	Cards	RBM wins	EMD wins	Ties	RBM zero-err k
3-rank	12	7	0	1	2
4-rank	16	7	0	1	2
5-rank	20	7	0	1	2
Total		21	0	3	

Table 4: Online learner convergence on 2-rank Rhode Island Hold'em. With random community cards, the learner discovers the natural cluster structure by game 100 and achieves near-perfect cache hit rates by game 600.

Games	Clusters	Cache Hit Rate
100	3	95.0%
200	3	97.5%
400	3	98.8%
600	3	99.5%

6.5 Online Learner Convergence

We run the online self-play learner (Algorithm 1) on 2-rank Rhode Island Hold'em with $\varepsilon = 50$ and random community cards.

Table 4 shows rapid convergence: the learner discovers all 3 natural clusters by game 100 (from 100 games with random community cards), and cache hit rate reaches 99.5% by game 600. With fixed community cards, convergence is even faster (same 3 clusters by game 50), confirming that the smaller strategic space is easier to learn.

6.6 Distance Memoization

The recursive structure of RBM creates many redundant subtree comparisons. Memoizing subtree distances yields significant speedups.

On an 8-tree benchmark (2-rank deck, 28 pairwise comparisons), memoized distance computation achieves a **94 $\times$ speedup** over the naive implementation, with a memo cache hit rate of approximately 97%. This demonstrates that game trees share substantial subtree structure that memoization exploits.

6.7 Full Limit Hold'em: Preflop Abstraction

We scale to full 2-player Limit Hold'em with a standard 52-card deck [Bowling et al., 2015]. After suit isomorphism, there are 50 canonical preflop hands. We compare RBM and EMD abstraction quality across 5 compression levels by measuring the mean EV error of each clustering.

Table 5 shows that **RBM wins 5–0** across all compression levels. The advantage is largest at $k = 25$ (42% less error), where RBM’s structural sensitivity matters most: it correctly separates hands that have similar equity distributions but different continuation tree structures (e.g., suited

Table 5: RBM vs EMD: preflop abstraction quality on full Limit Hold’em (50 canonical hands, 52-card deck). Mean EV error at each cluster count. Lower is better. **RBM wins all 5 levels.**

Clusters (k)	RBM Error	EMD Error	RBM Advantage	Winner
25	0.25	0.43	42% less error	RBM
15	0.37	0.43	14% less error	RBM
10	0.40	0.51	22% less error	RBM
5	0.60	0.69	13% less error	RBM
3	0.62	0.71	13% less error	RBM

Table 6: MCCFR bot tournament on full Limit Hold’em (20,000 hands, position-alternated). All trained bots beat both baselines. The RBM bot beats the EMD bot head-to-head, though the margin is not statistically significant (see text).

Matchup	Result (bb/hand)
RBM bot vs EMD bot	+0.02 (RBM wins) [†]
RBM bot vs Random	+1.24
EMD bot vs Random	+1.12
RBM bot vs Always-Call	+0.30
EMD bot vs Always-Call	+0.11

connectors vs. offsuit broadways). Even at extreme compression ($k = 3$), RBM maintains a 13% advantage.

6.8 Full Limit Hold’em: MCCFR Head-to-Head

We train MCCFR (Monte Carlo Counterfactual Regret Minimization) bots using RBM-based and EMD-based preflop abstractions, then evaluate in a 20,000-hand position-alternated match.

The head-to-head margin in Table 6 is +0.02 bb/hand—directionally consistent with the abstraction-quality results in Table 5, but well within sampling noise. We do *not* claim that better abstraction quality translated to stronger play in this experiment: at $n = 20,000$ hands the 95% CI of approximately $[-0.19, +0.23]$ comfortably covers zero. The same caveat applies to the differences against the random and always-call baselines, where both bots win by similar margins (+1.24 vs. +1.12 vs. random; +0.30 vs. +0.11 vs. always-call) but the differences are consistent with noise. A larger- n rerun, or a Slumbot-style benchmark match against a fixed strong opponent (cf. Johanson et al. 2013), would be the appropriate follow-up.

[†]*Statistical note:* The +0.02 bb/hand head-to-head margin is not statistically significant. With $\sigma \approx 15$ bb/hand for Limit Hold’em, $\text{SE}(20,000) \approx 0.11$ bb/hand, giving a 95% CI of approximately $[-0.19, +0.23]$ bb/hand. The directional advantage is consistent with the abstraction quality results (Table 5) but a larger sample is needed to confirm statistical significance.

6.9 Full Limit Hold’em: Online Learner

The online learner (Algorithm 1) scales to full Limit Hold’em, achieving a 99.4% cache hit rate— $60\times$ faster than offline distance computation. Approximately 400 postflop clusters emerge naturally with $\varepsilon = 0.5$, without any hand-crafted bucketing. This demonstrates that the RBM distance discovers meaningful strategic groupings even in the full game.

Table 7: Game tree sizes across poker variants (2-player heads-up unless noted). No-Limit games are substantially larger due to variable bet sizing, despite shallower stack depths.

Variant	Nodes	Notes
Limit HU	9,476	Fixed bet sizes
NL-HU 20bb	2,988	Short-stack push/fold
NL-HU 200bb	186,174	Deep-stack play
NL 6-max 20bb	~13M	Multi-player

Table 8: RBM vs EMD: preflop abstraction quality on NL-HU 20bb (50 canonical hands). Max EV error at each cluster count. Lower is better. **RBM wins 5–0** across all compression levels.

k	RBM Error	EMD Error	Improvement	Winner
25	0.11	0.41	73% less error	RBM
15	0.14	0.38	63% less error	RBM
10	0.25	0.38	34% less error	RBM
5	0.44	0.49	10% less error	RBM
3	0.45	0.48	5% less error	RBM

6.10 No-Limit Hold'em: Preflop Abstraction

We extend the evaluation to heads-up No-Limit Hold'em at two stack depths: 20bb (short-stack) and 200bb (deep-stack), the game variants underlying modern superhuman agents [Brown and Sandholm, 2018, Moravčík et al., 2017]. No-Limit introduces variable bet sizes that create qualitatively different game trees from Limit poker (Table 7).

Table 8 shows the 20bb results. **RBM wins 5–0** across all compression levels, with the largest advantage at fine-grained compression (73% less error at $k=25$, 63% at $k=15$).

Table 9 shows the 200bb deep-stack results. RBM dominates at fine-grained compression (60% less error at $k=25$, 40% at $k=15$) but loses at coarser compression ($k \leq 10$). Two notes are in order. First, the EMD advantage at coarse compression reflects a limitation of the experimental setup: the showdown-distribution trees used for tractable pairwise RBM distance (~ 161 nodes) do not fully capture the structural variation of the 186,174-node game tree. The 200bb results in this table should therefore be read as a lower bound on what full-tree RBM would achieve at this stack depth—the headline 5–0 framing we use elsewhere does *not* apply here, and we do not claim it. Second, the non-monotonic pattern (RBM wins at fine k , loses at coarse k) is consistent with the abstraction-pathology phenomenon documented by Waugh et al. [2009]: a more refined abstraction can produce a worse-playing strategy, depending on how the coarse-graining interacts with the underlying game’s strategic structure. We flag this rather than wave it away, because it is the kind of pattern an exploitability-aware evaluation (Johanson et al. 2013) is best suited to disentangle.

6.11 No-Limit Hold'em: MCCFR Head-to-Head

We train MCCFR bots on both NL-HU 20bb and 200bb using RBM and EMD preflop abstractions, then evaluate in 20,000-hand position-alternated matches.

Table 10 shows that the EMD bot wins both head-to-head matches. At 20bb, the margin is small (-0.12 bb/hand) and not statistically significant: with $\sigma \approx 10$ bb/hand for 20bb push/fold,

Table 9: RBM vs EMD: preflop abstraction quality on NL-HU 200bb (50 canonical hands, show-down distribution trees). Max EV error at each cluster count. Lower is better. RBM dominates at fine-grained compression; EMD wins at coarse levels.

k	RBM Error	EMD Error	Improvement	Winner
25	0.17	0.43	60% less error	RBM
15	0.28	0.46	40% less error	RBM
10	0.60	0.42	—	EMD
5	0.62	0.57	—	EMD
3	0.62	0.58	—	EMD

Table 10: MCCFR head-to-head on NL-HU at two stack depths (20,000 hands each, position-alternated, 50K training iterations). EMD wins at both depths, but neither result is statistically significant: the 20bb margin is within noise (95% CI includes zero), and the 200bb result is confounded by severe undertraining (~ 1 M info sets with only 50K iterations).

Depth	Result (bb/hand)	Info Sets	Training
20bb	-0.12 (EMD wins)	~ 70 K	adequate
200bb	-1.05 (EMD wins)	~ 1 M	undertrained

$SE(20,000) \approx 0.07$ bb/hand, giving a 95% CI of approximately $[-0.26, +0.02]$ bb/hand—marginal, with the interval nearly including zero. The result is consistent with the push/fold nature of short-stack play where raw equity is a strong predictor of optimal play.

At 200bb, the margin is larger (-1.05 bb/hand) but is not statistically reliable due to severe undertraining: the 200bb game produces ~ 1 M info sets per player, and with only 50K MCCFR iterations both bots are *severely undertrained*—average utility was still fluctuating at training end, far from convergence. The high variance of undertrained strategies makes the head-to-head outcome unreliable regardless of sample size. The 200bb game requires 500K+ iterations for meaningful head-to-head comparison. The abstraction quality results (Table 9), which do not depend on MCCFR training, are more reliable indicators of metric quality.

The key insight from Table 11 is that RBM’s structural advantage is clearest at fine-grained compression across both stack depths: 73% less error at $k=25$ for 20bb and 60% for 200bb. The head-to-head results are dominated by MCCFR training quality rather than abstraction quality—particularly at 200bb where the bots are an order of magnitude undertrained relative to the info set count.

7 Related Work

Tree edit distance. Zhang and Jiang [1994] and Zhang [1996] established that unrestricted edit distance on unordered labeled trees is MAX SNP-hard, motivating constrained variants computable in polynomial time. Kuboyama [2007] developed tractable variations using maximum weighted bipartite matching—essentially the same core operation as RBM. These works establish that matching-based tree distances are tractable and metrizable but do not consider game-theoretic applications.

Optimal transport on trees. Mémoli [2011] defines distances between metric measure spaces

Table 11: Cross-depth comparison of RBM vs EMD on No-Limit Hold’em. RBM’s abstraction quality advantage is consistent at fine-grained compression across both stack depths.

Depth	Nodes	Abstraction	Head-to-Head	Notes
20bb	2,988	RBM 5–0	EMD +0.12 bb/h	n.s. ($p > 0.05$)
200bb	186,174	RBM 2–3	EMD +1.05 bb/h	undertrained, n.s.

using optimal couplings; RBM can be viewed as a discrete, tree-structured specialization of the Gromov-Wasserstein framework. The Tree Mover’s Distance [Chuang et al., 2022] extends EMD to multisets of trees for GNN analysis. Yamada et al. [2018] formulate EMD on rooted labeled unordered trees built from complete subtrees. These works share the recursive optimal transport flavor but target different applications.

Game abstraction in poker. Gilpin and Sandholm [2007] introduced automated abstraction using k -means clustering with EMD on hand-strength histograms—flat abstraction that clusters on scalar distributions at each decision point. Ganzfried and Sandholm [2014] extended this with potential-aware EMD accounting for future card draws. Kroer and Sandholm [2014, 2018] provide the most theoretically rigorous bounds on exploitability introduced by abstraction, proving that level-by-level distance metrics control solution quality. Their bound is the natural target for any future exploitability-tightening of Theorem 3.2 (cf. Remark 3.3); RBM operates recursively through the full subtree rather than level-by-level, which is a strictly more expressive specification but does not on its own deliver a tighter exploitability bound.

Methodology of abstraction evaluation. Johanson et al. [2013] establish the standard methodology for measuring abstraction-induced exploitability in heads-up Limit Hold’em, and Waugh et al. [2009] document the *abstraction pathology* phenomenon—a more refined abstraction can produce a worse-playing strategy. Our results in Section 6.10 include exactly this pattern at 200bb (RBM wins at fine k , loses at coarse k), which we now discuss explicitly rather than wave away. A direct exploitability evaluation along the lines of Johanson et al. [2013] is the right next step for this work; the present evaluation reports abstraction-quality EV-error and head-to-head bb/hand only.

Online learning in games. CFR [Zinkevich et al., 2007] and its variants (CFR+, Monte Carlo CFR) achieve regret bounds for extensive-form games. AlphaZero [Silver et al., 2018] uses self-play with neural function approximation but lacks formal error bounds on the learned abstraction. MCTS [Coulom, 2006, Kocsis and Szepesvári, 2006] samples continuations to estimate node values—our online learner samples continuations to estimate node *identity* (location in the compressed graph). Lipschitz bandits [Kleinberg, 2005, Bubeck et al., 2011] exploit metric structure for faster learning; our setting derives the metric from game structure rather than assuming it.

The gap filled by this work. The existing literature has (1) matching-based tree distances with metric proofs, and (2) game tree abstraction with exploitability bounds. What was missing is the bridge: using a full recursive tree distance as the abstraction metric, proving that the distance bounds EV error, identifying the location problem as the fundamental complexity barrier, and the online learning formulation that dissolves it.

8 Conclusion

We have introduced the recursive bipartite matching distance, a metric on game trees that captures structural similarity—not just expected values, but the shape of the decision landscape. The metric induces a natural compression (the EV graph) with provable EV error bounds, and an online learner that builds this compression incrementally through self-play with bounded regret.

The results span three domains at multiple scales and report a consistent picture on *abstraction quality*, evaluated by EV error: lower error than EMD across the non-trivial compression levels of the Rhode Island Hold'em toy, lower error at all five tested compression levels on full Limit Hold'em (up to 42% reduction), and lower error at all five levels on No-Limit Hold'em 20bb (up to 73% reduction). On 200bb deep-stack No-Limit, RBM dominates only at fine-grained compression ($k \geq 15$) and loses at coarser levels—an asymmetry consistent with the abstraction-pathology phenomenon [Waugh et al., 2009] and amplified by our use of showdown-distribution trees as a tractable proxy for the 186,174-node game tree.

Gameplay results are more cautious. The MCCFR head-to-head matches we ran are not statistically distinguishable from zero (Limit Hold'em: +0.02 bb/h, $n=20,000$, 95% CI roughly $[-0.19, +0.23]$; NL 20bb and 200bb: small or noisy margins, the latter confounded by 50K-iteration undertraining of a $\sim 1\text{M}$ -info-set abstraction). We therefore do *not* claim that the abstraction quality advantage we measure has been demonstrated to translate to stronger play; the cleaner statement is that RBM yields demonstrably better abstractions by the EV-error metric we use, and that whether this carries over to a measurable head-to-head advantage requires more hands and adequately-trained MCCFR than we report here. We also do not yet report exploitability, the gold-standard evaluation established by Johanson et al. [2013]; closing that gap is the clearest follow-up.

Several directions for future work remain. *Exploitability evaluation along Johanson et al. [2013]*: the most important missing piece is a direct exploitability measurement of RBM-induced strategies, which would (a) test whether the abstraction-quality EV-error advantage predicts a tighter exploitability bound, and (b) detect any abstraction-pathology behaviour [Waugh et al., 2009] that the EV-error metric does not surface. *Tighter EV-to-exploitability bridge*: extending Theorem 3.2 into a Kroer–Sandholm-style exploitability bound (cf. Remark 3.3) is the natural theoretical follow-up. *Adequate training at 200bb*: with $\sim 1\text{M}$ info sets, the 200bb game needs 500K+ MCCFR iterations for a non-noise head-to-head read. *Full-tree RBM*: the showdown distribution trees we use for tractable 200bb pairwise RBM distance (~ 161 nodes) lose discriminative power compared to the full 186,174-node game trees; approximate matching via Sinkhorn distances [Cuturi, 2013] or learned embeddings could enable RBM on the full trees while preserving error bounds. *Multi-player*: the implementation supports 2–10 players, and 6-max 20bb already produces $\sim 13\text{M}$ -node game trees where equity-based clustering is increasingly inadequate. *Beyond poker*: the framework applies to any extensive-form game with tree-structured state spaces, including Stratego, Magic: The Gathering, and negotiation games. *Opponent modelling*: building an RBM-based EV graph of an opponent's play patterns could enable principled exploitation.

Reproducibility. All code, training scripts, and experiment configurations are available at the project repository (see title footnote). For each table in Section 6, the corresponding experiment binary under `bin/` produces the reported numbers from a fixed seed on a single CPU; the only stochasticity is in the deal sampler, which respects the per-experiment seed. We list MCCFR hyperparameters (sampling scheme, ε for ε -greedy exploration, regret-storage type, freeze cadence) inline with each table where they affect the result, and the full configuration is available in `rust/rbm_mccfr/src/config.rs` and `lib/cfr.ml`. Runs reported with confidence intervals are

single-seed; multi-seed CIs and bootstrap intervals on abstraction-quality tables are queued as part of the exploitability-evaluation follow-up.

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References

- Peter Auer, Nicolò Cesa-Bianchi, and Paul Fischer. Finite-time analysis of the multiarmed bandit problem. *Machine Learning*, 47(2-3):235–256, 2002.
- Michael Bowling, Neil Burch, Michael Johanson, and Oskari Tammelin. Heads-up limit hold’em poker is solved. *Science*, 347(6218):145–149, 2015.
- Noam Brown and Tuomas Sandholm. Superhuman AI for heads-up no-limit poker: Libratus beats top professionals. *Science*, 359(6374):418–424, 2018.
- Sébastien Bubeck, Rémi Munos, Gilles Stoltz, and Csaba Szepesvári. $\mathcal{X}$ -armed bandits. *Journal of Machine Learning Research*, 12:1655–1695, 2011.
- Ching-Yao Chuang, Stefanie Jegelka, and David Alvarez-Melis. Tree Mover’s distance: Bridging graph metrics and stability of graph neural networks. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2022.
- Rémi Coulom. Efficient selectivity and backup operators in Monte-Carlo tree search. In *Computers and Games*, pages 72–83, 2006.
- Marco Cuturi. Sinkhorn distances: Lightspeed computation of optimal transport. In *Advances in Neural Information Processing Systems (NeurIPS)*, pages 2292–2300, 2013.
- Sam Ganzfried and Tuomas Sandholm. Potential-aware imperfect-recall abstraction with earth mover’s distance in imperfect-information games. In *Proceedings of AAAI*, 2014.
- Andrew Gilpin and Tuomas Sandholm. Optimal Rhode Island Hold’em poker. In *Proceedings of AAMAS Workshop on Game Theoretic and Decision Theoretic Agents*, 2005.
- Andrew Gilpin and Tuomas Sandholm. Lossless abstraction of imperfect information games. *Journal of the ACM*, 54(5), 2007.
- Robert Kleinberg. Nearly tight bounds for the continuum-armed bandit problem. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2005.
- Michael Johanson, Neil Burch, Richard Valenzano, and Michael Bowling. Evaluating state-space abstractions in extensive-form games. In *Proceedings of AAMAS*, 2013.
- Levente Kocsis and Csaba Szepesvári. Bandit based Monte-Carlo planning. In *European Conference on Machine Learning*, pages 282–293, 2006.
- Christian Kroer and Tuomas Sandholm. Extensive-form game abstraction with bounds. In *Proceedings of EC*, 2014.
- Christian Kroer and Tuomas Sandholm. A unified framework for extensive-form game abstraction with bounds. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2018.

- Tetsuji Kuboyama. *Matching and Learning in Trees*. PhD thesis, University of Tokyo, 2007.
- Harold W. Kuhn. The Hungarian method for the assignment problem. *Naval Research Logistics Quarterly*, 2(1-2):83–97, 1955.
- Facundo Mémoli. Gromov-Wasserstein distances and the metric approach to object matching. *Foundations of Computational Mathematics*, 11(4):417–487, 2011.
- Matej Moravčík, Martin Schmid, Neil Burch, Viliam Lisý, Dustin Morrill, Nolan Bard, Trevor Davis, Kevin Waugh, Michael Johanson, and Michael Bowling. DeepStack: Expert-level artificial intelligence in heads-up no-limit poker. *Science*, 356(6337):508–513, 2017.
- David Silver, Thomas Hubert, Julian Schrittwieser, Ioannis Antonoglou, Matthew Lai, Arthur Guez, Marc Lanctot, Laurent Sifre, Dhharshan Kumaran, Thore Graepel, Timothy Lillicrap, Karen Simonyan, and Demis Hassabis. A general reinforcement learning algorithm that masters chess, shogi, and Go through self-play. *Science*, 362(6419):1140–1144, 2018.
- Kevin Waugh, David Schnizlein, Michael Bowling, and Duane Szafron. Abstraction pathologies in extensive games. In *Proceedings of AAMAS*, 2009.
- Eizo Yamada, Kaiichiro Ota, and Kazuhisa Makino. Earth mover’s distance on rooted labeled unordered trees. In *Proceedings of COCOA*, pages 420–434, 2018.
- Kaizhong Zhang. A constrained edit distance between unordered labeled trees. *Algorithmica*, 15(3):205–222, 1996.
- Kaizhong Zhang and Tao Jiang. Some MAX SNP-hard results concerning unordered labeled trees. *Information Processing Letters*, 49(5):249–254, 1994.
- Martin Zinkevich, Michael Johanson, Michael Bowling, and Carmelo Piccione. Regret minimization in games with incomplete information. In *Advances in Neural Information Processing Systems (NeurIPS)*, pages 1729–1736, 2007.

A Full Proof: RBM Distance Is a Metric (Theorem 2.2)

Theorem (Restated). *Let $\mathcal{T}$ be the space of rooted trees with real-valued leaves. Define $d : \mathcal{T} \times \mathcal{T} \rightarrow \mathbb{R}_{\geq 0}$ as in Definition 2.1, with leaf distance d_L and phantom penalty δ . Suppose:*

1. d_L is a metric on the leaf value space.
2. $\delta(S) \geq 0$ for all subtrees S .
3. **Phantom triangle property:** for any subtrees S_1, S_2 , $d(S_1, S_2) \leq \delta(S_1) + \delta(S_2)$.

Then d is a metric on $\mathcal{T}$.

Proof. We verify each axiom by structural induction on the maximum depth k of the trees involved. The inductive hypothesis is that d restricted to trees of depth $< k$ satisfies all three metric axioms. The base case ($k = 0$, both trees are leaves) is immediate because d reduces to d_L , which is a metric by assumption.

Axiom 1: Identity of indiscernibles — $d(T, T) = 0$. Let T have children $\{c_1, \dots, c_m\}$. The cost matrix W for matching T against itself has entries $W[i, j] = d(\text{sub}(c_i), \text{sub}(c_j))$. The child sets are identical and have the same cardinality, so no phantom padding is needed. The identity matching $M^* = \{(c_i, c_i)\}_{i=1}^m$ has cost:

$$\text{cost}(M^*) = \sum_{i=1}^m d(\text{sub}(c_i), \text{sub}(c_i)) = 0,$$

where each term vanishes by the inductive hypothesis. Since the Hungarian method finds the minimum-cost matching and $\text{cost}(M^*) = 0$ with all costs nonneg, $d(T, T) = 0$.

Conversely, if $d(T_1, T_2) = 0$, then every matched pair (c_1^i, c_2^j) has $d(\text{sub}(c_1^i), \text{sub}(c_2^j)) = 0$ (nonneg terms summing to zero), and there are no phantom-matched children (since $\delta(S) \geq 0$ and any phantom match would force the sum above zero, unless $\delta(S) = 0$, corresponding to an empty subtree). By induction, each matched pair of subtrees is identical, so $T_1 = T_2$. $\square$

Axiom 2: Symmetry — $d(T_1, T_2) = d(T_2, T_1)$. The cost matrix W' for $d(T_2, T_1)$ is the transpose of W for $d(T_1, T_2)$:

- Real-real entries: $d(\text{sub}(c_2^j), \text{sub}(c_1^i)) = d(\text{sub}(c_1^i), \text{sub}(c_2^j))$ by inductive symmetry.
- Phantom entries: δ depends only on the real subtree, not which side it appears on.

The minimum-cost perfect matching is invariant under transposition. $\square$

Axiom 3: Triangle inequality — $d(T_1, T_3) \leq d(T_1, T_2) + d(T_2, T_3)$. Fix three trees T_1, T_2, T_3 with children $\{a_1, \dots, a_p\}$, $\{b_1, \dots, b_q\}$, $\{c_1, \dots, c_r\}$. Let M_{12} achieve cost $d(T_1, T_2)$ and M_{23} achieve cost $d(T_2, T_3)$.

Step 1: Construct a feasible matching M_{13} . Compose through T_2 . Each b_j appears once in M_{12} and once in M_{23} . This creates:

- *Through-matched:* $a_i \xrightarrow{M_{12}} b_j \xrightarrow{M_{23}} c_k$. Assign $(a_i, c_k) \in M_{13}$.
- *Left-dangling:* $a_i \rightarrow b_j$ but $b_j \rightarrow \phi$. Assign $a_i \rightarrow \phi$ in M_{13} .
- *Right-dangling:* $\phi \rightarrow b_j \rightarrow c_k$. Assign $c_k \rightarrow \phi$ in M_{13} .
- *Both-phantom:* $a_i \rightarrow \phi$ in M_{12} (not through any b_j). Assign $a_i \rightarrow \phi$ in M_{13} . Symmetrically for $\phi \rightarrow c_k$ in M_{23} .

Step 2: Bound cost. For through-matched pairs, by the inductive triangle inequality:

$$d(\text{sub}(a_i), \text{sub}(c_k)) \leq d(\text{sub}(a_i), \text{sub}(b_j)) + d(\text{sub}(b_j), \text{sub}(c_k)).$$

For left-dangling a_i (via b_j matched to phantom in M_{23}): the cost in M_{13} is $\delta(\text{sub}(a_i))$. The corresponding costs in M_{12} and M_{23} are $d(\text{sub}(a_i), \text{sub}(b_j))$ and $\delta(\text{sub}(b_j))$. The 1-Lipschitz condition $\delta(\text{sub}(a_i)) \leq d(\text{sub}(a_i), \text{sub}(b_j)) + \delta(\text{sub}(b_j))$ ensures the bound holds. Right-dangling and both-phantom cases are symmetric.

Step 3: Sum. Every edge of M_{12} and M_{23} appears exactly once on the right-hand side:

$$\text{cost}(M_{13}) \leq \text{cost}(M_{12}) + \text{cost}(M_{23}) = d(T_1, T_2) + d(T_2, T_3).$$

Since $d(T_1, T_3) \leq \text{cost}(M_{13})$ by optimality of the Hungarian matching:

$$d(T_1, T_3) \leq d(T_1, T_2) + d(T_2, T_3). \quad \square$$

□

Remark A.1. With $\delta(S) = d(S, \emptyset)$, the phantom triangle property and 1-Lipschitz condition are both instances of the triangle inequality: $d(S_1, S_2) \leq d(S_1, \emptyset) + d(\emptyset, S_2) = \delta(S_1) + \delta(S_2)$ and $d(S_1, \emptyset) \leq d(S_1, S_2) + d(S_2, \emptyset)$. The theorem is self-contained.

B Full Proof: EV Error Bound (Theorem 3.2)

Theorem (Restated). Let T_1, T_2 be rooted trees with real-valued leaves, $d(T_1, T_2) = \varepsilon$. Let $T^* = \text{merge}(T_1, T_2)$ with equal-weight leaf averaging. Then $|\text{EV}(T^*) - \text{EV}(T_i)| \leq \varepsilon/2$ for $i \in \{1, 2\}$.

Proof. By structural induction on tree depth.

Base case: leaves. $T_1 = \text{leaf}(v_1)$, $T_2 = \text{leaf}(v_2)$, $d(T_1, T_2) = |v_1 - v_2| = \varepsilon$. The merge is $T^* = \text{leaf}((v_1 + v_2)/2)$, so:

$$|\text{EV}(T^*) - \text{EV}(T_1)| = \left| \frac{v_1 + v_2}{2} - v_1 \right| = \frac{|v_2 - v_1|}{2} = \frac{\varepsilon}{2}. \quad \checkmark$$

Inductive case: internal nodes. Let T_1 have children $\{a_1, \dots, a_m\}$ and T_2 have children $\{b_1, \dots, b_n\}$ with $m \leq n$. Let M^* be the optimal matching with matched-pair costs $d_i = d(\text{sub}(a_i), \text{sub}(b_{\sigma(i)}))$ and phantom costs $\delta(b_j)$ for $j \notin \text{im}(\sigma)$:

$$\varepsilon = \sum_{i=1}^m d_i + \sum_{j \notin \text{im}(\sigma)} \delta(b_j).$$

Case 1: Phantom children dropped (lossy merge). With uniform weighting:

$$\begin{aligned} |\text{EV}(T^*) - \text{EV}(T_1)| &= \left| \frac{1}{m} \sum_{i=1}^m \text{EV}(c_i^*) - \frac{1}{m} \sum_{i=1}^m \text{EV}(\text{sub}(a_i)) \right| \\ &\leq \frac{1}{m} \sum_{i=1}^m |\text{EV}(c_i^*) - \text{EV}(\text{sub}(a_i))| \\ &\leq \frac{1}{m} \sum_{i=1}^m \frac{d_i}{2} \\ &= \frac{1}{2m} \sum_{i=1}^m d_i \leq \frac{\varepsilon}{2}, \end{aligned}$$

where the third line uses the inductive hypothesis on each matched pair, and the last step uses $\sum_i d_i \leq \varepsilon$.

Equal-structure case ($m = n$, **no phantoms**). The bound tightens:

$$|\text{EV}(T^*) - \text{EV}(T_1)| \leq \frac{1}{2m} \sum_{i=1}^m d_i = \frac{\varepsilon}{2m}.$$

For $m \geq 2$, this is at most $\varepsilon/4$, strictly better than the stated bound.

The bound $\varepsilon/2$ is tight only at leaves ($m = 0$) or degenerate single-child chains ($m = 1$ at every internal node). $\square$

Corollary (Weighted Merges). *For weights w_1, w_2 with $w_1 + w_2 = 1$, merged leaf values are $v^* = w_1 v_1 + w_2 v_2$. The error bound generalizes to:*

$$|\text{EV}(T^*) - \text{EV}(T_i)| \leq w_{3-i} \cdot \varepsilon.$$

For $w_1 = w_2 = 1/2$, this recovers $\varepsilon/2$. For $w_1 \rightarrow 1$, the error for T_1 vanishes while that for T_2 approaches ε .

C Full Proof: Online Regret Bound (Theorem 5.1)

Theorem (Restated). *Consider the online EV graph learner with threshold $\varepsilon > 0$, playing T games against a stationary environment. Let K be the number of strategic equivalence classes and $V_{\max} = \max_t |\text{EV}(T_t)|$. Then:*

1. *Exploration bound: total cache misses $\leq K$.*
2. *Per-step exploitation error: at most $\varepsilon/2$.*
3. *Cumulative regret: $R(T) \leq KV_{\max} + (T - K)\varepsilon/2$.*
4. *Average regret: $R(T)/T \rightarrow \varepsilon/2$ as $T \rightarrow \infty$.*
5. *With $\varepsilon(t) = \varepsilon_0/\sqrt{t}$: $R(T) = O(\sqrt{T})$.*

Proof. Part 1: Exploration bound. Define a strategic equivalence class as a maximal set $\{T : d(T, T') \leq \varepsilon\}$ centered at some T' . Since the game is finite, the number of such classes under fixed $\varepsilon > 0$ is some finite K .

A cache miss occurs when T_t lies outside all existing clusters: $d(T_t, \text{rep}(c)) > \varepsilon$ for every cluster c . A new cluster is created with T_t as representative.

Each equivalence class generates at most one miss: once $T_t \in \mathcal{C}$ triggers a miss, the new cluster covers all future $T_{t'} \in \mathcal{C}$ since $d(T_{t'}, T_t) \leq \varepsilon$ by definition. Therefore total misses $\leq K$. $\square$

Part 2: Per-step exploitation error. When exploiting cluster c^* on game tree T_t with $d(T_t, \text{rep}(c^*)) \leq \varepsilon$, the cluster representative serves the role of the merged tree. By Theorem 3.2:

$$r_t = |\text{EV}(\text{rep}(c^*)) - \text{EV}(T_t)| \leq \frac{d(T_t, \text{rep}(c^*))}{2} \leq \frac{\varepsilon}{2}.$$

Part 3: Cumulative regret bound. Partition the T games into exploration games $\mathcal{E}$ (cache misses, $|\mathcal{E}| \leq K$) and exploitation games $\mathcal{X}$ (cache hits, $|\mathcal{X}| \geq T - K$).

Exploration: worst-case per-game regret is $V_{\max}$. Summing: $\sum_{t \in \mathcal{E}} r_t \leq KV_{\max}$.

Exploitation: per-game regret $\leq \varepsilon/2$. Summing: $\sum_{t \in \mathcal{X}} r_t \leq (T - K)\varepsilon/2$.

Total:

$$R(T) = \sum_{t \in \mathcal{E}} r_t + \sum_{t \in \mathcal{X}} r_t \leq KV_{\max} + (T - K)\frac{\varepsilon}{2}.$$

Part 4: Average regret convergence.

$$\frac{R(T)}{T} \leq \frac{KV_{\max}}{T} + \frac{(T-K)}{T} \cdot \frac{\varepsilon}{2} \xrightarrow{T \rightarrow \infty} \frac{\varepsilon}{2}.$$

The exploration cost $KV_{\max}$ is a one-time overhead amortized over all future games. After all K classes are discovered, every subsequent game is pure exploitation with per-game regret at most $\varepsilon/2$. $\square$

Part 5: Sublinear regret via ε -annealing. Let $\varepsilon(t) = \varepsilon_0/\sqrt{t}$. Let $K(\varepsilon)$ denote the number of equivalence classes at threshold ε , bounded by N (total distinct game trees).

Total exploration events across all T rounds is at most $K(\varepsilon(T)) \leq N$, since each decrease in ε that splits a class triggers at most one new exploration event.

Exploitation regret at time t is at most $\varepsilon(t)/2 = \varepsilon_0/(2\sqrt{t})$. Summing:

$$\sum_{t \in \mathcal{X}} \frac{\varepsilon_0}{2\sqrt{t}} \leq \sum_{t=1}^T \frac{\varepsilon_0}{2\sqrt{t}} \leq \frac{\varepsilon_0}{2} \cdot 2\sqrt{T} = \varepsilon_0\sqrt{T},$$

using the standard bound $\sum_{t=1}^T t^{-1/2} \leq 2\sqrt{T}$.

Adding exploration cost:

$$R(T) \leq NV_{\max} + \varepsilon_0\sqrt{T} = O(\sqrt{T}).$$

Average regret:

$$\frac{R(T)}{T} \leq \frac{NV_{\max}}{T} + \frac{\varepsilon_0}{\sqrt{T}} \rightarrow 0 \quad \text{as } T \rightarrow \infty. \quad \square$$

Remark C.1 (Comparison with Standard Bandit Bounds). *The $O(\sqrt{T})$ rate under ε -annealing matches the minimax-optimal rate for multi-armed bandits with K arms [Auer et al., 2002]. The EV graph learner is effectively a bandit over clusters, with the RBM distance providing a geometry on the arm space: nearby arms have similar payoffs by the EV error bound. This is analogous to Lipschitz bandits [Kleinberg, 2005], where metric structure enables faster learning. In the EV graph setting, the metric is not assumed but derived from the game structure.*

Remark C.2 (Tightness). *The exploration term $KV_{\max}$ is tight when each class is first encountered at a worst-case game with adversarial exploration (zero payoff). In practice, heuristic strategies achieve reasonable EV, so the effective exploration cost is much lower. The exploitation term $(T-K)\varepsilon/2$ is tight at leaves; for trees with branching factor $m > 1$, Theorem 3.2 gives a tighter per-step bound of $\varepsilon/(2m)$.*